

Testing and Calibration of Gas Sensors

Key words

Calibration, test, validation, gas sensors, gas measuring instruments, characteristic curves, ageing, reproducibility, cross sensitivity

Quantities and items tested

Determination of gas mixture composition (in particular hydrogen-, ammonia-, methane-, sulfur dioxide and ethanol mixtures), characteristic curves of gas sensors, gas measuring devices

Testing range

Uncertainty of results

Volume fraction of gas mixture components $1 \cdot 10^{-6} \text{ m}^3/\text{m}^3$ to $0.99 \text{ m}^3/\text{m}^3$	from	2 %	to	5 %
Humidity 0.05 % to 99 % relative humidity	from	0.5 %	to	2 %
Temperature -40 °C to +180 °C	from	0.1 K	to	0.3 K
Impedance 10 mΩ to 100 MΩ	from	0.1 Ω	to	50 Ω
Resistance 10 mΩ to 100 MΩ	from	0.1 Ω	to	50 Ω
Capacity 1 pF to 100 nF	from	0.1 pF	to	2.5 pF

Fields of application

Calibration, testing and validation of gas sensors and gas measuring instruments

Methodology and instrumentation

Generation and determination of gas mixtures with defined composition, complex gas mixing and analysis system

Qualification and quality assurance

The accuracy of the procedure is regularly verified using certified reference gas mixtures.

The measuring equipment and reference standards of the entire system are subject to a rigorous calibration program providing traceability to the SI.

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Further information

General

Gas sensors are calibrated and tested using a test system where test gas mixtures of defined composition are generated dynamically from appropriate parent gases in cylinders. These test gas mixtures are transferred into test chambers containing the sensors under investigation. The gas blending system provides for continuous variation of mixture composition, including humidification, at a high dynamic range (left part of figure). Gas blending is performed using mass-flow controllers (MFC), which control four different gas streams. The blending process and the resulting composition are regulated by varying the gas flow through the MFCs. The system is able to generate gas mixtures containing up to four components, an inert carrier gas (synthetic air or nitrogen) and humidity. Two test chambers have been developed, optimised for a small and a larger volume, respectively, and accurately defined gas flow. These test chambers are equipped with five (or up to ten) sensors of different types and provide for measurement of gas temperature between $-40\text{ }^{\circ}\text{C}$ and $+180\text{ }^{\circ}\text{C}$. The gas mixtures generated are analysed using a dew-point mirror and a quadrupole mass-spectrometer to check the accuracy of the pre-determined mixture composition and humidity. The humidification of the gas mixture is based on the saturation method using a glass bubbler. The humidified carrier gas is cooled down in a gas cooler to adjust the dew point. Finally, the humidified carrier gas is mixed with the test gas, and dry inert carrier gas is added. A personal computer controls all parts of the system via an IEEE-bus net. The right part of the figure shows a schematic plan of the measuring system.



Operating principle

